Casual Document for Quantitative Mixing Time Results for Single Site Gibbs in Stochastic Block Model

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Chapter 1

The Stochastic Block Model and the Single Site Gibbs Method

The procedure of analyzing graphical data using Bayesian stochastic block model with single site Gibbs procedure seems to have been first mentioned in Kemp et al. (2004) and slightly later in Xu et al. (2006). In this chapter, we describe the simplest setup of both the model and the single site Gibbs method by following a tutorial article Schmidt and Mørup (2013). We will fill in the missing steps and justify that the proposed method is indeed the standard single site systematic scan Gibbs method.

1.1 The Bayesian Stochastic Block Model

For the moment, let us consider the simplest case where the graphical data received by the statistician is just one simple graph of $N \geqslant 3$ vertices labeled as $1, 2, \ldots, N$, represented by its adjacency matrix $X \in \{0,1\}^{N \times N}$ where $X_{i,j} = 1$ if and only if there is an edge between vertex i and vertex j. In particular, we have $X_{i,i} = 0$ for all i and $X_{i,j} = X_{j,i}$ for all i, j. If we want to, we can consider later the case which our data contains multiple graphs of the same size generated independently identically distributed from a distribution in the same model, but we shall not do so in this section.

To describe the model, some notations have to be introduced. Let us write Θ for the collection of all partitions of the set $\{1, 2, \ldots, N\}$. That is, each $z \in \Theta$ is a collection of nonempty disjoint subsets of $\{1, \ldots, N\}$ whose union equals to $\{1, \ldots, N\}$. Very frequently, we shall present an element $z \in \Theta$ by writing it as a vector $z = (z_1, z_2, \ldots, z_N) \in \{1, \ldots, N\}^N$ where z_i denotes the block to which vertex i belongs. For example, if $N = 5$, both $z = (1, 2, 1, 3, 3)$ and $z = (5, 1, 5, 2, 2)$ correspond to the partition $z = \{\{1, 3\}, \{2\}, \{4, 5\}\}$. z is said to be a *restricted growth string* if the blocks are ordered as they first appear: $z_1 \equiv 1$ and, for every $i \in \{2, 3, \ldots, N\}$, we have

$$z_i \in \left\{1, 2, \ldots, 1 + \max_{j<i} z_j\right\}.$$

In the previous example, $(1, 2, 1, 3, 3)$ is a restricted growth string and the other is not.

We assume that X is generated from a distribution in the (standard) stochastic block model (SBM) with unknown number of blocks. The model contains two parameters: the partition z ranged over Θ and the connectivity probabilities $\phi_{k,l} \in [0, 1]$, where $k \in z$ and $l \in z$ are the communities. The model assumes that, given z and ϕ, independently over all other edges, an edge is joined between a vertex in block k and a vertex in block l with probability ϕ_{kl}. In symbols:

$$p(X \mid z, \phi) = \prod_{1 \leqslant i < j \leqslant N} \phi_{z_i, z_j}^{x_{ij}} (1 - \phi_{z_i, z_j})^{1 - x_{ij}}, \tag{1.1}$$

where we have written $p\left(X \mid z, \phi\right)$ in place of $p\left((X_{i,j})_{1 \leqslant i < j \leqslant N} \mid z, \phi\right)$. Another look at (1.1) supply a second formula:

$$p\left(X \mid z, \phi\right) = \prod_{k,l \in z} \phi_{k,l}^{m_{k,l}} (1 - \phi_{k,l})^{\bar{m}_{k,l}}, \tag{1.2}$$

where $m_{k,l}(\bar{m}_{k,l})$ is the number of edges(non-edges) between block k and block l, and, here and below, $\prod_{k,l \in z}$ runs over unordered pairs $\{k, l\}$ of blocks, including $k = l$.

In order to do Bayesian statistics, priors need to be posed for z and for ϕ. The standard practice for this model, as discussed in Schmidt and Mørup (2013), is to first implement the conjugate prior for ϕ, integrate ϕ out to get a formula for $p\left(X \mid z\right)$, treat the model as if z is the only parameter, give z a prior distribution, and then carry out Markov chain Monte Carlo(MCMC) on the posterior distribution $p(z \mid X) \propto p(X \mid z)p(z)$. This is the procedure we will follow.

With hyperparameters $a > 0$ and $b > 0$ fixed by the statistician, we require all $\phi_{k,l}$'s to have a prior distribution of Beta(a, b), independently with respect to each other and over all other variables. Consequently:

$$\begin{aligned} p(X \mid z) &= \int_{[0,1]^{\frac{|z|(|z|+1)}{2}}} p(X \mid z, \phi) p(\phi) \mathrm{d}\phi = \int_{[0,1]^{\frac{|z|(|z|+1)}{2}}} \prod_{k,l \in z} \frac{1}{B(a,b)} \phi_{k,l}^{m_{k,l}+a-1} (1 - \phi_{k,l})^{\bar{m}_{k,l}+b-1} \mathrm{d}\phi \\ &= \prod_{k,l \in z} \frac{1}{B(a,b)} \int_0^1 \phi_{k,l}^{m_{k,l}+a-1} (1 - \phi_{k,l})^{\bar{m}_{k,l}+b-1} \mathrm{d}\phi_{k,l} \\ &= \prod_{k,l \in z} \frac{B(m_{k,l} + a, \bar{m}_{k,l} + b)}{B(a,b)}, \end{aligned} \tag{1.3}$$

where $B(\cdot, \cdot)$ is the beta function. From now on, we treat z as the only parameter of the model, defining a density of X via (1.3). As justified in Schmidt and Mørup (2013), a commonly used prior distribution for z is known as the *Chinese restaurant process (CRP)*. With a hyperparameter $A > 0$ fixed by the statistician, CRP(A) is a probability measure on Θ defined as follows: with restricted growth notation, $z_1 \equiv 1$ and, for every $i \in \{2, 3, \ldots, N\}$, given $z_1, z_2, \ldots, z_{i-1}$ who have already used a total of $K^{(i-1)}$ blocks, z_i will join an existing block $k \in \{1, 2, \ldots, K^{(i-1)}\}$ with probability

$$\frac{n_k^{(i-1)}}{A + i - 1},$$

where $n_k^{(i-1)}$ is the number of vertices among $1, 2, \ldots, i-1$ who have been assigned to block k, and will create a new block $K^{(i-1)} + 1$ with probability

$$\frac{A}{A + i - 1}.$$

CRP(A) does have a clean joint density (with respect to the counting measure on Θ)

$$p(z) = \frac{A^{|z|}\Gamma(A)}{\Gamma(A + N)} \prod_{k \in z} \Gamma(n_k), \tag{1.4}$$

where n_k is the number of vertices assigned to block k according to z, and $\Gamma(\cdot)$ is the gamma function.

By Bayes's theorem, according to (1.3) and (1.4), the posterior distribution of z given X is

$$p(z \mid X) \propto p(X \mid z)p(z) \propto \left[\prod_{k,l \in z} \frac{B(m_{k,l} + a, \bar{m}_{k,l} + b)}{B(a,b)}\right] \left[A^{|z|} \prod_{k \in z} \Gamma(n_k)\right]. \tag{1.5}$$

We remark that the $B(a, b)$ term in the denominator should not go to the normalizing constant: it gets multiplied by $|z|(|z| + 1)/2$ times.

1.2 The Single Site Systematic Scan Gibbs Suggested in Schmidt and Mørup (2013)

We present in this section the single site systematic scan Gibbs method suggested in Schmidt and Mørup (2013). The method produces a sequence of partitions $z^{(0)}, z^{(1)}, z^{(2)}, \ldots \in \Theta$, where $z^{(0)}$ is an initial partition and $z^{(t)}$ is the partition obtained after t sweeps. In each sweep, the vertices $1, 2, \ldots, N$ are visited in this fixed order, and the visited vertex is reassigned to a block while the blocks of all the other vertices are kept fixed.

To describe one such reassignment, let $z \in \Theta$ be the current partition and let $i \in \{1, \ldots, N\}$ be the vertex being visited. We write $z^{\setminus i}$ for the partition of $\{1, \ldots, N\} \setminus \{i\}$ obtained from z by removing vertex i from its block (and discarding this block if it becomes empty). For blocks $k, l \in z^{\setminus i}$, we write

- $n_k^{\setminus i}$ for the number of vertices in block k;
- $r_{i,k} = \sum_{j \in k} X_{i,j}$ for the number of edges between vertex i and block k;
- $m_{k,l}^{\setminus i}(\bar{m}_{k,l}^{\setminus i})$ for the number of edges(non-edges) between block k and block l, not counting vertex i, that is,

$$m_{k,l}^{\setminus i} = \begin{cases} \displaystyle\sum_{j \in k} \sum_{j' \in l} X_{j,j'}, & k \neq l, \\ \displaystyle\sum_{j,j' \in k,\, j<j'} X_{j,j'}, & k = l, \end{cases} \qquad \bar{m}_{k,l}^{\setminus i} = \begin{cases} n_k^{\setminus i} n_l^{\setminus i} - m_{k,l}^{\setminus i}, & k \neq l, \\ \frac{1}{2} n_k^{\setminus i}(n_k^{\setminus i} - 1) - m_{k,l}^{\setminus i}, & k = l. \end{cases}$$

The vertex i is then either put into an existing block $k \in z^{\setminus i}$ or put into a new block $\{i\}$ of its own. (In particular, if $\{i\}$ is a block of z, keeping vertex i alone counts as the second option.) The choice of joining the existing block $k \in z^{\setminus i}$ is given the unnormalized weight

$$w_k = n_k^{\setminus i} \prod_{l \in z^{\setminus i}} \frac{B(m_{k,l}^{\setminus i} + r_{i,l} + a,\ \bar{m}_{k,l}^{\setminus i} + n_l^{\setminus i} - r_{i,l} + b)}{B(m_{k,l}^{\setminus i} + a,\ \bar{m}_{k,l}^{\setminus i} + b)}, \tag{1.6}$$

and the choice of creating a new block is given the unnormalized weight

$$w_{\text{new}} = A \prod_{l \in z^{\setminus i}} \frac{B(r_{i,l} + a,\ n_l^{\setminus i} - r_{i,l} + b)}{B(a, b)}. \tag{1.7}$$

The block of vertex i is then drawn from the categorical distribution proportional to these weights: vertex i joins block $k \in z^{\setminus i}$ with probability w_k / W and creates a new block with probability w_{new}/W, where $W = w_{\text{new}} + \sum_{k \in z^{\setminus i}} w_k$. In Schmidt and Mørup (2013), this categorical distribution is presented as the conditional distribution of the block of vertex i given the blocks of all the other vertices and X.

The full procedure reads as follows.

1. **Initialization.** Choose an initial partition $z^{(0)} \in \Theta$ and a total number $2T$ of sweeps. In the implementation of Schmidt and Mørup (2013), all vertices start in one single block, that is, $z^{(0)} = \{\{1, \ldots, N\}\}$.
2. **Sweeps.** For $t = 1, 2, \ldots, 2T$: set $z = z^{(t-1)}$ and, for $i = 1, 2, \ldots, N$ in turn,
   (a) compute $z^{\setminus i}$ and, for all $k, l \in z^{\setminus i}$, the quantities $n_l^{\setminus i}$, $r_{i,l}$, $m_{k,l}^{\setminus i}$ and $\bar{m}_{k,l}^{\setminus i}$ from the current z;
   (b) compute the weights w_k, $k \in z^{\setminus i}$, by (1.6) and w_{new} by (1.7);
   (c) draw the block of vertex i from the categorical distribution described above;
   (d) replace z by the resulting partition, namely $z^{\setminus i}$ with block k replaced by $k \cup \{i\}$ if block k is drawn, and $z^{\setminus i} \cup \{\{i\}\}$ if a new block is drawn.

After vertex N has been visited, set $z^{(t)} = z$.

3. **Output.** Discard the first half $z^{(1)}, \dots, z^{(T)}$ as burn-in and use $z^{(T+1)}, \dots, z^{(2T)}$ as approximate samples from the posterior distribution (1.5).

In the implementation of Schmidt and Mørup (2013), the weights are computed on the logarithmic scale (via the logarithm of the beta function), and the largest log weight is subtracted from all log weights before exponentiating and normalizing.

1.3 Justifications

We now justify that the method presented in Section 1.2 indeed corresponds to the single site systematic scan Gibbs sampler in a slightly different space.

Classically speaking, the single site systematic scan Gibbs sampler refers to a solution of the following problem: suppose that we wish to sample from a probability measure $\pi(\theta)$ on a σ-finite space (Θ, F_Θ, μ) defined by a density $\mathrm{d}\pi(\theta) = p(\theta)\mathrm{d}\mu(\theta)$. If it happens that Θ is a product space (i.e. $\Theta = \Theta_1 \times \cdots \times \Theta_N$ and $\mu = \mu_1 \times \cdots \times \mu_N$) and the full conditional distributions $p(\theta_i \mid \theta_{(-i)})$, where $\theta_{(-i)} = (\theta_1, \dots, \theta_{i-1}, \theta_{i+1}, \dots, \theta_N)$, can be sampled, then the *single site systematic scan Gibbs sampler* samples π by iteratively sampling at each coordinate and standard theory covers its qualitative convergence results Robert and Casella (2004).

The problem here is that, when Θ is the collection of all partitions of $\{1, \dots, N\}$, it is entirely unclear how Θ can be written as a product space $\prod_{i=1}^N \Theta_i$, where Θ_i corresponds to the vertex i. If we simply represent each $z \in \Theta$ using the restricted growth notation, then the vector $z_{(-i)}$ obtained by simply cropping the ith entry is no longer in the restricted growth notation and further analysis based on this notation is invalid.

(Here is a clean but slightly annoying solution I was able to find.)

Instead of working on Θ, we work on an augmented version of it. Let us define:

$$\tilde{\Theta} = \prod_{i=1}^N \tilde{\Theta}_i = \{1, \dots, N\}^N,$$

$F_{\tilde{\Theta}} = 2^{\tilde{\Theta}}$, and μ_i be the counting measure on $\tilde{\Theta}_i = \{1, \dots, N\}$. For each $c = (c_1, \dots, c_N) \in \tilde{\Theta}$ and $i \in \{1, \dots, N\}$, let $c_{(-i)} = (c_1, \dots, c_{i-1}, c_{i+1}, \dots, c_N)$ really be the vector cropping out the ith coordinate. For each $l \in \{1, \dots, N\}$, let us write $(l, c_{(-i)})$ for the vector $(c_1, \dots, c_{i-1}, l, c_{i+1}, \dots, c_N)$, filling l to the ith place. A map $\Pi : \tilde{\Theta} \to \Theta$ can be defined by setting $\Pi(c)$ to be the partition which c represents, in the way discussed as in Section 1.1. If $p(z)$ is any density on Θ (with respect to the counting measure), then

$$\tilde{p}(c) = \frac{(N - |\Pi(c)|)!}{N!} p(\Pi(c))$$

is a density with respect to μ on $(\tilde{\Theta}, F_{\tilde{\Theta}}, \mu)$ with $C \sim \tilde{p}$ implying $\Pi(C) \sim p$.

Let us reformulate the model with parameter space being the augmented space. In this way, the model is no longer identifiable and (1.3) now reads

$$\tilde{p}(X \mid c) = \prod_{k,l \in \Pi(c)} \frac{B(m_{k,l} + a, \bar{m}_{k,l} + b)}{B(a, b)}. \tag{1.8}$$

The prior (1.4) now becomes

$$\tilde{p}(c) = \frac{(N - |\Pi(c)|)!}{N!} \frac{A^{|\Pi(c)|}\Gamma(A)}{\Gamma(A + N)} \prod_{k \in \Pi(c)} \Gamma(n_k). \tag{1.9}$$

The posterior (1.5) now becomes

$$\tilde{p}(c \mid X) \propto \frac{(N-|\Pi(c)|)!}{N!}\left[\prod_{k,l\in\Pi(c)}\frac{B(m_{k,l}+a,\bar{m}_{k,l}+b)}{B(a,b)}\right]\left[A^{|\Pi(c)|}\prod_{k\in\Pi(c)}\Gamma(n_k)\right]. \tag{1.10}$$

We now proceed to compute the full conditional probabilities. Given the clear product space structure, we separate $c=(c_i,c_{(-i)})$ without worry. We have, for each $l\in\{1,\dots,N\}$:

$$\tilde{p}(c_i=l\mid c_{(-i)},X)=\frac{\tilde{p}(X\mid c_i=l,c_{(-i)})\tilde{p}(c_i=l\mid c_{(-i)})\tilde{p}(c_{(-i)})}{\tilde{p}(c_{(-i)},X)}\propto\tilde{p}(X\mid c_i=l,c_{(-i)})\tilde{p}(c_i=l\mid c_{(-i)}). \tag{1.11}$$

Therefore, the only thing left is to compute these two terms.

To compute $\tilde{p}(c_i=l\mid c_{(-i)})$, we use the joint distribution $\tilde{p}(l,c_{(-i)})$ given in (1.9), which requires some notations. Let us write $\Pi(c_{(-i)})$ for the partition of the set $\{1,\dots,i-1,i+1,\dots,N\}$ obtained by starting with $\Pi(c)$ and deleting vertex i (if the action creates an empty block, delete the empty block as well) and, for each $k\in\Pi(c_{(-i)})$, $n_k^{(-i)}$ for the number of vertices in block k. It is clear that $\sum_{k\in\Pi(c_{(-i)})}n_k^{(-i)}=N-1$ because the vertex i is not in any $k\in\Pi(c_{(-i)})$. We now discuss by cases. If the choice $c_i=l$ creates a new block from $\Pi(c_{(-i)})$ (that is, $\left|\Pi(l,c_{(-i)})\right|=\left|\Pi(c_{(-i)})\right|+1$), then

$$\tilde{p}(l,c_{(-i)})=\frac{(N-\left|\Pi(c_{(-i)})\right|-1)!}{N!}\frac{A^{|\Pi(c_{(-i)})|+1}\Gamma(A)}{\Gamma(A+N)}\times\Gamma(1)\times\prod_{k\in\Pi(c_{(-i)})}\Gamma(n_k^{(-i)})=\frac{A}{N-\left|\Pi(c_{(-i)})\right|}q(c_{(-i)}),$$

where

$$q(c_{(-i)})=\frac{(N-\left|\Pi(c_{(-i)})\right|)!}{N!}\frac{A^{|\Pi(c_{(-i)})|}\Gamma(A)}{\Gamma(A+N)}\prod_{k\in\Pi(c_{(-i)})}\Gamma(n_k^{(-i)}).$$

Secondly, suppose that the choice $c_i=l$ enters an existing block from $\Pi(c_{(-i)})$, say inside a block $r\in\Pi(c_{(-i)})$. In this case, we have $\left|\Pi(l,c_{(-i)})\right|=\left|\Pi(c_{(-i)})\right|$,

$$\tilde{p}(l,c_{(-i)})=\frac{(N-\left|\Pi(c_{(-i)})\right|)!}{N!}\frac{A^{|\Pi(c_{(-i)})|}\Gamma(A)}{\Gamma(A+N)}\times\Gamma(n_r^{(-i)}+1)\times\prod_{k\in\Pi(c_{(-i)}),k\neq r}\Gamma(n_k^{(-i)})=n_r^{(-i)}q(c_{(-i)}).$$

We can thus combine the two cases and observe that, in the formula of $\tilde{p}(l,c_{(-i)})$, the block which l belongs will determine what is multiplied to $q(c_{(-i)})$ and this is the only thing that depends on l. We thus conclude that

$$\tilde{p}(c_i=l\mid c_{(-i)})=\frac{1}{A+N-1}\begin{cases}n_r^{(-i)} & l\text{ enters block }r\in\Pi(c_{(-i)}),\\ \frac{A}{N-\left|\Pi(c_{(-i)})\right|} & l\text{ creates a new block.}\end{cases} \tag{1.12}$$

It can be verified directly that this is a proper density:

$$\sum_{l=1}^{N}\tilde{p}(c_i=l\mid c_{(-i)})=\frac{1}{A+N-1}\left(\left(N-\left|\Pi(c_{(-i)})\right|\right)\frac{A}{N-\left|\Pi(c_{(-i)})\right|}+\sum_{r\in\Pi(c_{(-i)})}n_r^{(-i)}\right)=\frac{A+N-1}{A+N-1}=1.$$

To compute $\tilde{p}(X\mid c_i=l,c_{(-i)})$, we use (1.8) and again discuss by cases. We introduce a bit more notations. For each $s,t\in\Pi(l,c_{(-i)})$, let us write $m_{s,t}(l,c_{(-i)})(\bar{m}_{s,t}(l,c_{(-i)}))$ for the number of edges(non-edges) between block s and block t in the graph X when vertex i is assigned to block l. Also, let $m_{s,t}(c_{(-i)})(\bar{m}_{s,t}(c_{(-i)}))$ be the

number of edges(non-edges) between block s and block t in the graph X when vertex i is deleted (along with edges connecting i to other vertices), which is 0 if block s or block t becomes empty. If the choice $c_i = l$ creates a new block (called just l) from $\Pi(c_{(-i)})$, then (1.8) reads (the factor for the pair $\{l, l\}$ equals $B(a,b)/B(a,b) = 1$ and is omitted)

$$\begin{aligned}
&\tilde{p}(X \mid c_i = l, c_{(-i)}) \\
&= \left[\prod_{s,t\in\Pi(c_{(-i)})} \frac{B(m_{s,t}(l, c_{(-i)}) + a, \bar{m}_{s,t}(l, c_{(-i)}) + b)}{B(a,b)}\right] \left[\prod_{s\in\Pi(c_{(-i)})} \frac{B(m_{s,l}(l, c_{(-i)}) + a, \bar{m}_{s,l}(l, c_{(-i)}) + b)}{B(a,b)}\right] \\
&= \left[\prod_{s,t\in\Pi(c_{(-i)})} \frac{B(m_{s,t}(c_{(-i)}) + a, \bar{m}_{s,t}(c_{(-i)}) + b)}{B(a,b)}\right] \\
&\left[\prod_{s\in\Pi(c_{(-i)})} \frac{B(m_{s,l}(c_{(-i)}) + e_s^i + a, \bar{m}_{s,l}(c_{(-i)}) + n_s^{(-i)} - e_s^i + b)}{B(a,b)}\right] \\
&= \left[\prod_{s,t\in\Pi(c_{(-i)})} \frac{B(m_{s,t}(c_{(-i)}) + a, \bar{m}_{s,t}(c_{(-i)}) + b)}{B(a,b)}\right] \left[\prod_{s\in\Pi(c_{(-i)})} \frac{B(e_s^i + a, n_s^{(-i)} - e_s^i + b)}{B(a,b)}\right],
\end{aligned}$$

where e_s^i is the number of edges between vertex i and block s in the graph X and $n_s^{(-i)}$ is the number of vertices in block s without the vertex i. In the last equality, we have used the fact that $m_{s,l}(c_{(-i)}) = \bar{m}_{s,l}(c_{(-i)}) = 0$ because block l is empty once vertex i is deleted. Secondly, suppose that the choice $c_i = l$ enters an existing block from $\Pi(c_{(-i)})$, say inside a block $r \in \Pi(c_{(-i)})$, then (1.8) reads

$$\begin{aligned}
&\tilde{p}(X \mid c_i = l, c_{(-i)}) \\
&= \left[\prod_{s,t\in\Pi(c_{(-i)}), s\neq r, t\neq r} \frac{B(m_{s,t}(l, c_{(-i)}) + a, \bar{m}_{s,t}(l, c_{(-i)}) + b)}{B(a,b)}\right] \\
&\left[\prod_{s\in\Pi(c_{(-i)})} \frac{B(m_{s,r}(l, c_{(-i)}) + a, \bar{m}_{s,r}(l, c_{(-i)}) + b)}{B(a,b)}\right] \\
&= \left[\prod_{s,t\in\Pi(c_{(-i)}), s\neq r, t\neq r} \frac{B(m_{s,t}(c_{(-i)}) + a, \bar{m}_{s,t}(c_{(-i)}) + b)}{B(a,b)}\right] \\
&\left[\prod_{s\in\Pi(c_{(-i)})} \frac{B(m_{s,r}(c_{(-i)}) + e_s^i + a, \bar{m}_{s,r}(c_{(-i)}) + n_s^{(-i)} - e_s^i + b)}{B(a,b)}\right].
\end{aligned}$$

We can thus combine the two cases (block l being r in the second case) to conclude that

$$\tilde{p}(X \mid c_i = l, c_{(-i)}) = L(c_{(-i)}) \prod_{s\in\Pi(c_{(-i)})} \frac{B(m_{s,l}(c_{(-i)}) + e_s^i + a, \bar{m}_{s,l}(c_{(-i)}) + n_s^{(-i)} - e_s^i + b)}{B(m_{s,l}(c_{(-i)}) + a, \bar{m}_{s,l}(c_{(-i)}) + b)}, \tag{1.13}$$

where

$$L(c_{(-i)}) = \prod_{s,t\in\Pi(c_{(-i)})} \frac{B(m_{s,t}(c_{(-i)}) + a, \bar{m}_{s,t}(c_{(-i)}) + b)}{B(a,b)}$$

does not depend on l.

Combining (1.11), (1.12) and (1.13), we see that

$$\tilde{p}(c_i = l \mid c_{(-i)}, X) \propto \begin{cases} n_r^{(-i)} \displaystyle\prod_{s \in \Pi(c_{(-i)})} \frac{B(m_{s,r}(c_{(-i)}) + e_s^i + a, \bar{m}_{s,r}(c_{(-i)}) + n_s^{(-i)} - e_s^i + b)}{B(m_{s,r}(c_{(-i)}) + a, \bar{m}_{s,r}(c_{(-i)}) + b)}, & l \text{ enters block } r \in \Pi(c_{(-i)}), \\ \dfrac{A}{N - \left|\Pi(c_{(-i)})\right|} \displaystyle\prod_{s \in \Pi(c_{(-i)})} \frac{B(e_s^i + a, n_s^{(-i)} - e_s^i + b)}{B(a, b)}, & l \text{ creates a new block.} \end{cases}$$

1.4 Next Steps

To proceed

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